Turing Pi 2.5 ARM64 K3s Cluster

Benchmark Comparison vs. Typical x86 K3s

Storage • Control Plane • CIS Security

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Version 1.0

1. Introduction

This document compares the benchmark results from a Turing Pi 2.5 ARM64 K3s cluster against typical x86-based K3s deployments. The goal is to provide context for the ARM64 results and help administrators understand where the Turing Pi cluster excels, where it faces limitations, and how it compares to traditional server hardware running the same workloads.

All ARM64 results were collected firsthand from the Turing Pi 2.5 cluster. The x86 reference figures are drawn from published benchmarks, community reports, and vendor documentation for comparable K3s deployments, as cited throughout the document.

2. Hardware Comparison

Specification	Turing Pi 2.5 (ARM64)	Typical x86 K3s
Nodes	4 (3 server + 1 worker)	3–5 (bare-metal or VM)
CPU per Node	RK3588: 4×A76 + 4×A55(8 cores @ 2.4/1.8 GHz)	Intel Xeon / AMD EPYC(4–16 cores @ 2.5–3.5 GHz)
RAM per Node	32 GB LPDDR4x	16–64 GB DDR4/DDR5 ECC
Storage per Node	1 TB NVMe (PCIe 3.0 ×4)	500 GB–2 TB NVMe (PCIe 3.0/4.0)
Network (Internal)	1 Gbps (BMC backplane)	10 GbE / 25 GbE
Form Factor	Mini-ITX (4 modules)	1U–2U rack / tower / VM
TDP per Node	~5–10 W	~65–150 W
Total Power Draw	~30–50 W	~300–800 W
Approx. Cost	~$900–$1,100 complete	~$3,000–$10,000+

The most significant hardware difference is network bandwidth. The Turing Pi 2.5 uses a 1 Gbps BMC backplane for inter-node traffic, while typical x86 K3s clusters use 10 GbE or faster. This 10× network gap is the primary bottleneck for replicated storage operations.

3. Storage Benchmark Comparison

3.1 Raw NVMe Performance (Local-Path)

The local-path storage class bypasses network replication, testing raw NVMe performance through the Kubernetes storage stack. This isolates the CPU and NVMe controller performance differences between ARM64 and x86.

Test	Turing Pi (RK3588)	Typical x86 (Xeon)	x86 Advantage
Random Read IOPS (4K)	148,000	300,000–500,000	2–3.4×

Random Write IOPS (4K)	31,600	80,000–150,000	2.5–4.7×
Sequential Read (MB/s)	3,021	3,000–7,000	1–2.3×
Sequential Write (MB/s)	2,943	2,000–5,000	0.7–1.7×

Analysis: The RK3588's PCIe 3.0 ×4 NVMe interface delivers impressive sequential throughput—actually matching or approaching entry-level x86 NVMe performance. The gap is most visible in random IOPS, where x86 CPUs benefit from higher clock speeds, deeper out-of-order execution, and more mature NVMe driver optimizations. However, the Turing Pi's raw NVMe numbers are far beyond what most containerized workloads will saturate.

3.2 Longhorn Replicated Storage (3 Replicas)

This is where the 1 Gbps network bottleneck becomes the dominant factor. Longhorn synchronously replicates writes to all replicas before acknowledging the I/O, meaning every write must traverse the network.

Test	Turing Pi (1 GbE)	Typical x86 (10 GbE)	x86 Advantage
Random Read IOPS (4K)	7,797	15,000–25,000	1.9–3.2×
Random Write IOPS (4K)	4,397	6,000–19,000	1.4–4.3×
Sequential Read (MB/s)	135	400–900	3–6.7×
Sequential Write (MB/s)	51	200–500	3.9–9.8×
Mixed R/W IOPS (75/25)	6,213 / 2,082	12,000 / 4,000	~2×

Analysis: Sequential write at 51 MB/s confirms the 1 Gbps network ceiling (~125 MB/s theoretical, ~100 MB/s practical, halved by 3-replica synchronous write). This is the Turing Pi's most significant limitation. An x86 cluster with 10 GbE networking delivers 4–10× higher Longhorn throughput. The gap narrows for random IOPS because small I/O operations are less bandwidth-bound and more latency-bound.

Key Insight: The Turing Pi's Longhorn IOPS (7,797 read / 4,397 write) are still adequate for most home lab and edge workloads—databases like PostgreSQL, monitoring stacks, and media servers will run well. The bottleneck primarily affects bulk data operations like large backups or high-throughput streaming.

3.3 Storage Overhead: Longhorn vs. Local-Path

Metric	Turing Pi (1 GbE)	Typical x86 (10 GbE)
Random Read IOPS Overhead	19× (148K → 7.8K)	6–10×
Random Write IOPS Overhead	7× (31.6K → 4.4K)	5–8×
Seq. Read Bandwidth Overhead	22× (3,021 → 135 MB/s)	4–8×

Seq. Write Bandwidth Overhead	58× (2,943 → 51 MB/s)	4–10×

The overhead ratios tell the real story. On x86 with 10 GbE, Longhorn typically costs 4–10× versus local storage. On the Turing Pi with 1 GbE, the overhead ranges from 7–58×. The extreme 58× penalty on sequential writes is almost entirely attributable to the 1 Gbps network. Upgrading to 2.5 GbE (via the Turing Pi 2.5's optional networking) or future 10 GbE would dramatically close this gap.

4. Control Plane Performance Comparison

Control plane benchmarks measure the Kubernetes API server's responsiveness for pod lifecycle operations. These tests are CPU and etcd-bound rather than network-bound, so the comparison is more directly about ARM64 vs. x86 processing power.

Operation	Turing Pi (RK3588)	Typical x86 (Xeon)	x86 Advantage
Deploy 25 pods (cold start)	28.7 s	10–20 s	1.4–2.9×
Scale 25→50 (cached)	3.2 s	1–3 s	1–3.2×
Scale 50→5	532 ms	200–500 ms	1–2.7×
Delete deployment	193 ms	50–200 ms	1–3.9×
Pod distribution	6–7 per node (balanced)	Similar (scheduler-dependent)	—

Analysis: The 28.7-second cold start includes container image pulls, which are heavily I/O and CPU-bound (decompression). Once images are cached, scaling 25 additional pods takes only 3.2 seconds—a 9× improvement and well within the range of x86 performance. The scale-down and delete operations at sub-second latency are effectively equivalent to x86.

Key Insight: For steady-state operations (scaling existing deployments, rolling updates with cached images), the RK3588 performs comparably to entry-level x86 servers. The gap is primarily in first-deployment scenarios where image decompression on the ARM64 cores takes longer than on a high-frequency Xeon. In practice, this is a one-time cost per image per node.

5. CIS Security Benchmark Comparison

The kube-bench CIS benchmark results are not performance-dependent—they test configuration compliance, not speed. The comparison here is about how K3s defaults compare across platforms.

Section	PASS	FAIL	WARN	INFO	x86 PASS	x86 FAIL
1. Control Plane	40	0	11	10	~40	0
2. Etcd	7	0	0	0	7	0
3. Control Plane Cfg	0	0	5	0	0	0–5
4. Worker Node	11	5*	2	5	11–16	0–5*
5. Policies	0	0	35	0	0	0
Total	58	5*	53	15	58–63	0–10*

** The 5 FAILs in Section 4 are false positives—K3s embeds kubelet settings internally rather than exposing them as CLI flags. The remediation text confirms the defaults are correct. This is identical on x86 K3s.*

Analysis: CIS benchmark results are effectively identical between ARM64 and x86 K3s installations. The same binary handles security configuration regardless of CPU architecture. The 5 Section 4 “failures” are a kube-bench detection limitation with K3s’s embedded architecture, not an actual security deficiency. The 35 Section 5 WARNs are manual policy reviews (RBAC, Pod Security Standards, NetworkPolicies) that require human judgment regardless of platform.

Etcd perfect score (7/7) deserves special mention—K3s’s embedded etcd is properly configured with TLS mutual authentication and unique certificate authorities on both ARM64 and x86.

6. Overall Assessment

6.1 Where the Turing Pi Competes Well

- **Raw NVMe throughput:** Sequential read/write bandwidth matches entry-level x86 NVMe, thanks to the RK3588's PCIe 3.0 ×4 interface. Most single-container workloads will never saturate this.
- **Control plane responsiveness:** Cached pod scaling (3.2s for 25 pods) and sub-second scale-down/delete are operationally equivalent to x86 for small-to-medium clusters.
- **Security posture:** Identical CIS compliance profile to x86 K3s. No ARM64-specific security gaps.
- **Power efficiency:** 30–50W total for a 4-node HA cluster vs. 300–800W for x86. This is a 10–20× advantage in performance-per-watt for workloads that don't need x86's peak throughput.
- **Cost:** ~$1,000 for a complete HA cluster with 128 GB total RAM and 4 TB NVMe vs. $3,000–$10,000+ for equivalent x86.

6.2 Where x86 Has Clear Advantages

- **Replicated storage throughput:** 10 GbE networking delivers 4–10× higher Longhorn bandwidth. This is the single biggest gap and is network-limited, not CPU-limited.
- **Random IOPS:** x86 CPUs with higher clock speeds and deeper pipelines deliver 2–4× higher random IOPS on local NVMe. This matters for database-heavy workloads.
- **Cold start image pull:** Container image decompression is faster on high-frequency x86 cores. First deployment of large images takes ~30s on ARM64 vs. ~10–20s on x86.
- **Software ecosystem:** Some container images still lack ARM64 builds, requiring workarounds (as encountered with yasker/kbench during benchmarking).

6.3 The Network Bottleneck

The 1 Gbps backplane is the single largest performance constraint on the Turing Pi 2.5. Nearly every metric where x86 shows a large advantage traces back to network bandwidth rather than CPU capability. If the Turing Pi cluster had 10 GbE networking, the Longhorn gap would shrink from 4–10× to approximately 1.5–3× (the residual CPU difference).

The Turing Pi 2.5 does support an optional 2.5 GbE networking module, which would provide a 2.5× improvement in network-bound operations. Future SBC clustering boards with 10 GbE would largely close the storage performance gap with x86.

7. Workload Recommendations

7.1 Ideal Workloads for Turing Pi

- Home lab Kubernetes learning and experimentation
- Edge computing and IoT gateway clusters
- Low-traffic web applications and APIs

- Monitoring stacks (Prometheus, Grafana, Loki)
- Media servers and home automation (Home Assistant, Plex)
- CI/CD runners for ARM64 builds
- Always-on services where power efficiency matters

7.2 Workloads Better Suited for x86

- High-throughput databases with heavy write loads
- Large-scale data processing (Spark, Kafka)
- Workloads requiring >100 MB/s sustained replicated writes
- Applications without ARM64 container images
- Latency-sensitive applications requiring sub-millisecond storage I/O

8. Conclusion

The Turing Pi 2.5 with RK3588 compute modules delivers remarkably capable Kubernetes performance for its power envelope and price point. Raw NVMe performance approaches entry-level x86, control plane operations are operationally comparable for small clusters, and security compliance is identical.

The primary limitation is the 1 Gbps network backplane, which constrains Longhorn replicated storage to 51 MB/s sequential write—adequate for most home lab and edge workloads, but a significant bottleneck compared to x86 clusters with 10 GbE. This is a platform limitation rather than an ARM64 limitation, and future networking upgrades would largely close the gap.

For a cluster that draws under 50W, costs around $1,000, fits on a desk, and runs a full HA K3s deployment with Longhorn, Grafana, and CIS-compliant security—the Turing Pi 2.5 offers exceptional value that no x86 platform can match at this power and price point.